

DSA0404 - Fundamentals of Data Science

1. What is Data Science? Explain its primary objective.

Definition:

Data Science is an interdisciplinary field that combines statistics, mathematics, computer science, artificial intelligence, and domain knowledge to collect, process, analyze, and interpret data in order to extract meaningful insights and support decision-making.

It uses techniques such as Machine Learning, Data Mining, Data Analysis, Data Visualization, and Big Data technologies to solve real-world problems.

Primary Objective

The primary objective of Data Science is to convert raw data into useful knowledge that helps organizations make better decisions, predict future trends, automate processes, and solve complex business problems.

Objectives include:

- Extract useful information from large datasets.
 - Discover hidden patterns and relationships.
 - Predict future outcomes.
 - Support data-driven decision making.
 - Improve efficiency and business performance.
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2. Why is Data Science important in today's digital world? Mention any four reasons.

Data Science has become essential because enormous amounts of data are generated every day through the internet, smartphones, social media, sensors, and business transactions.

Importance

1. Better Decision Making

Organizations use data to make accurate and informed business decisions instead of relying on guesswork.

2. Predictive Analysis

Helps predict future events such as customer demand, disease outbreaks, weather forecasting, and stock prices.

3. Business Growth

Companies improve sales, customer satisfaction, and operational efficiency using data-driven insights.

4. Automation

Machine Learning automates repetitive tasks such as fraud detection, recommendation systems, and customer support.

Other reasons:

- Personalized services
 - Healthcare improvements
 - Scientific research
 - Smart cities
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3. List any four benefits of Data Science in business and society.

Benefits

1. Improved Decision Making

Businesses analyze customer and market data to make better strategic decisions.

2. Cost Reduction

Identifies inefficient processes and optimizes operations, reducing costs.

3. Better Customer Experience

Recommendation systems provide personalized products and services.

4. Healthcare Improvements

Predicts diseases, assists diagnosis, and improves patient care.

Other benefits:

- Fraud detection
 - Better education
 - Traffic management
 - Environmental monitoring
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4. Mention four real-world applications of Data Science.

Applications

Healthcare

Disease prediction, medical imaging, drug discovery.

Finance

Fraud detection, credit scoring, stock market prediction.

E-commerce

Product recommendations, demand forecasting, customer segmentation.

Transportation

Traffic prediction, route optimization, autonomous vehicles.

Other applications:

- Agriculture
- Banking
- Sports analytics
- Manufacturing
- Social media

5. What are the different facets of data? Briefly explain each.

The major facets of data are:

1. Structured Data

Data stored in rows and columns.

Example:

Student database, Employee records.

2. Semi-Structured Data

Does not follow a strict table format but contains tags or keys.

Example:

XML, JSON.

3. Unstructured Data

Has no predefined format.

Example:

Videos, Images, Emails, Audio files.

4. Metadata

Data about data.

Example:

File size, creation date, author name.

5. Big Data

Large and complex datasets that traditional databases cannot efficiently process.

Example:

Social media posts, IoT sensor data.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured Data	Semi-Structured Data	Unstructured Data
Organized in rows and columns	Partially organized	No predefined format
Stored in relational databases	Stored using tags or keys	Stored as files
Easy to query	Moderately easy	Difficult to analyze
SQL is used	JSON/XML parsers	AI and NLP techniques
Example: Student database	Example: XML, JSON	Example: Images, Videos, Emails

7. What is Big Data? State its key characteristics (5 Vs).

Definition

Big Data refers to extremely large and complex datasets that cannot be processed efficiently using traditional database systems.

5 Vs of Big Data

1. Volume

Huge amount of data.

2. Velocity

Speed at which data is generated and processed.

3. Variety

Different types of data:

- Structured
- Semi-structured
- Unstructured

4. Veracity

Quality and reliability of data.

5. Value

Useful insights obtained from data.

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem consists of tools and technologies used for storing, processing, analyzing, and visualizing massive datasets.

Components

- Data Sources
- Data Storage
- Data Processing
- Data Analysis
- Data Visualization

Technologies

- Hadoop
- Spark
- Hive
- HBase

Other technologies:

- Kafka
 - Pig
 - Flume
 - Cassandra
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9. What are the major phases of the Data Science process?

The Data Science process consists of:

1. Problem Definition
 2. Data Collection
 3. Data Retrieval
 4. Data Cleaning
 5. Data Integration
 6. Data Transformation
 7. Exploratory Data Analysis (EDA)
 8. Model Building
 9. Model Evaluation
 10. Deployment
 11. Monitoring
 12. Presentation of Results
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10. What is data retrieval? Mention two common sources from which data can be retrieved.

Definition

Data retrieval is the process of collecting or accessing required data from various storage systems for analysis.

Common Sources

- Databases (MySQL, Oracle)
- APIs/Web Services

Other sources:

- CSV files
 - Cloud storage
 - Web scraping
 - IoT devices
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11. Why is data collection considered an important step in the Data Science process?

Data collection is important because the quality of analysis depends on the quality of collected data.

Importance

- Provides accurate information.
- Improves prediction accuracy.
- Reduces bias.
- Helps build reliable models.
- Supports better business decisions.

Quote:

"High-quality data leads to high-quality insights."

12. What is data cleansing? List any four common data-cleaning techniques.

Definition

Data cleansing is the process of detecting and correcting inaccurate, incomplete, duplicate, or inconsistent data.

Techniques

- Remove duplicate records.
- Handle missing values.
- Correct inconsistent formats.
- Detect and remove outliers.

Other techniques:

- Standardization
- Error correction

- Noise removal
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13. What is data integration? Why is it necessary in Data Science?

Definition

Data integration is the process of combining data from multiple sources into a single unified dataset.

Necessity

- Eliminates redundancy.
- Improves data consistency.
- Provides complete information.
- Enables better analysis.
- Supports accurate predictions.

Example:

Combining customer information from sales, marketing, and finance databases.

14. Explain data transformation with suitable examples.

Definition

Data transformation converts raw data into a suitable format for analysis.

Examples

- Converting date format:
10/07/2026 → 2026-07-10
- Scaling numerical values:
Income from ₹50,000 to 0.85 using normalization.
- Encoding categorical values:
Male → 1
Female → 0

Other transformations:

- Aggregation
- Log transformation
- Standardization

15. What is data analysis? Mention any four techniques used for data analysis.**Definition**

Data analysis is the process of examining, organizing, and interpreting data to discover useful information and support decision-making.

Techniques

- Statistical Analysis
- Regression Analysis
- Classification
- Clustering

Other techniques:

- Time Series Analysis
- Association Rule Mining
- Machine Learning

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive	Predictive	Prescriptive
Explains what happened	Predicts what will happen	Suggests what should be done
Uses historical data	Uses Machine Learning	Uses optimization techniques
Reports and dashboards	Forecasting	Decision support
Example: Monthly sales report	Sales prediction	Best pricing strategy

17. What is meant by building a model in Data Science? Mention the basic steps involved.**Definition**

Model building is the process of training a Machine Learning algorithm using historical data to make predictions or classifications.

Steps

1. Select algorithm.

2. Split data into training and testing sets.
3. Train the model.
4. Test the model.
5. Tune parameters.
6. Save and deploy the model.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Importance

Model evaluation measures how well a model performs on unseen data.

It helps to:

- Measure accuracy.
- Detect overfitting.
- Compare models.
- Improve performance.

Classification Metrics

- Accuracy
- Precision
- Recall
- F1-Score

Other metrics:

- ROC Curve
- AUC
- Confusion Matrix
- Specificity

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Importance

Presenting findings helps stakeholders understand results and make informed decisions.

Benefits:

- Simplifies complex data.
- Highlights trends.
- Supports decision-making.
- Improves communication.

Visualization Tools

- Tableau
- Microsoft Power BI
- Matplotlib
- Excel Charts

Other tools:

- Python (Seaborn, Plotly)
- Google Data Studio

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Model Deployment

After successful training and evaluation, models are deployed so users can access predictions in real time.

Deployment methods include:

- Web applications
- Mobile applications
- Cloud platforms
- REST APIs
- Embedded systems

Applications

1. Netflix movie recommendation.
2. Banking fraud detection.

3. Medical disease diagnosis.
4. Amazon product recommendation.

Other applications:

- Face recognition
- Spam email detection
- Self-driving cars
- Weather forecasting